



## ***AlphaServer DS20L System*** **Technical Summary**



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# hp AlphaServer DS20L Systems

The *HP AlphaServer DS20L*, a high-density rack-mount server, delivers supercomputing in tight spaces. Besides the high-performance technical computing market, the DS20L dual-processor system is ideal for high-throughput, front-end cluster ISP/Internet applications and telecom applications. Up to 40 of these 1-U systems can be mounted in a single cabinet.

## System Overview

The *AlphaServer DS20L* systems have two Alpha 21264B 833 MHz CPU chips. Up to 2 Gbytes of memory can be installed. Second-level cache is 4 Mbytes per processor. The switch-based system interconnect exploits the full potential of the Alpha chip.

The system measures 17.5 x 20.5 x 1.75 inches (1U) and can be rackmounted in a 79- or 67-inch M-series cabinet. In the 79-inch cabinet up to 40 systems can be installed, or if more storage is desired, there can be up to six StorageWorks shelves per cabinet. A rackmount slide kit is mandatory with each system in a cabinet.

Each system has two drive bays. One is for the system disk drive (18.2, 36.4, or 72.8 GB disk). The other can hold an optional CD-ROM.

Two 10/100 fast Ethernet controllers, an Ultra3 SCSI controller, and an IDE controller are integrated into the system. For expansion there are two 64-bit, 33 MHz PCI slots (2/3 length). Two serial ports and a bi-directional enhanced parallel port are also available.

Systems can be purchased with the *Tru64 UNIX* operating system installed, or they can be purchased without any operating system, allowing customers to install Linux.

Along with other *AlphaServer* products, the DS20L uses the 64-bit Alpha RISC architecture. The system can be ordered with the *Tru64 UNIX* operating system or without an operating system.

For more information on *AlphaServer DS20L* systems, see <http://www.compaq.com/alphaserver>.

## Features and Benefits

- **Package and Price**

The DS20L system provides what you need for the targeted applications at an attractive price.

- **Performance**

The Alpha chip when combined with a switch-based interconnect demonstrates its full potential. This switch-based system provides a memory bandwidth of up to 2.6 Gbytes/sec (peak) with a 256-bit memory bus running at 83 MHz.

- **Space Saver**

When floor space is limited, the DS20L systems allow you to pack maximum computes into a minimal footprint.

## Third-Generation Alpha Chip

The third generation of the Alpha microprocessor, the Alpha 21264B, is a superscalar superpipelined implementation of the Alpha architecture. The first offering of this chip was known as EV6; these systems are offered with the EV68 chip (.18 micron). Each chip has over 15.2 million transistors.

Designed for performance, the Alpha 21264B achieves this goal by carefully studied and simulated architectural and circuit analysis. The 21264B memory system also enables the high performance levels. On-chip and off-chip caches provide for very low latency data access, which allows for very high bandwidth data access. (The size of the off-chip cache is 4 MB per processor running at 278 MHz DDR.)

Internal to each chip is a 64-Kbyte instruction cache (I-cache) and a 64-Kbyte data cache (D-cache).

- **I-cache.** 64 Kbytes, two-way set-associative, virtually addressed cache with 64-byte blocks
- **D-cache.** 64 Kbytes, two-way set-associative, virtually indexed, physically tagged, writeback cache with 64-byte blocks

### Chip Operation

Several key design choices were made in the chip architecture to maximize performance: Four instructions are fetched each cycle, and then how those instructions are handled boosts the speed of execution. Register renaming assigns a unique storage location with each write reference to a register, avoiding register dependencies that can be a potential bottleneck to processor performance.

Another design feature, out-of-order execution, permits instructions to execute in an order different from the order that the instructions are fetched. In effect, instructions execute as soon as possible. This allows for faster execution since critical path computations are started and completed as soon as possible.

In addition, the Alpha 21264B employs speculative execution to maximize performance. It speculatively fetches and executes instructions even though it may not know immediately whether the instruction will be on the final execution path. This is particularly useful, for instance, when the 21264B predicts branch directions and speculatively executes down the predicted path. The sophisticated branch prediction in the 21264B coupled with the speculative and dynamic execution extracts the most instruction parallelism from applications.

For more information about the chip, see:

<http://www.compaq.com/alphaserver/download/ev6chip.pdf>

### Alpha 21264B Features

- Out-of-order instruction execution
- Large (64 Kbyte) on-chip data and instruction caches
- Improved branch prediction through intuitive execution
- Register renaming
- Increased bandwidth for high-speed access to second-level cache and system memory
- Motion video instructions
- Square root and divide instructions
- All instructions are 32 bits long and have a regular instruction format
- Floating-point unit, supports DIGITAL and IEEE floating-point data types
- 80 integer registers, 64 bits wide
- 72 floating-point registers, 64 bits wide

## Architecture

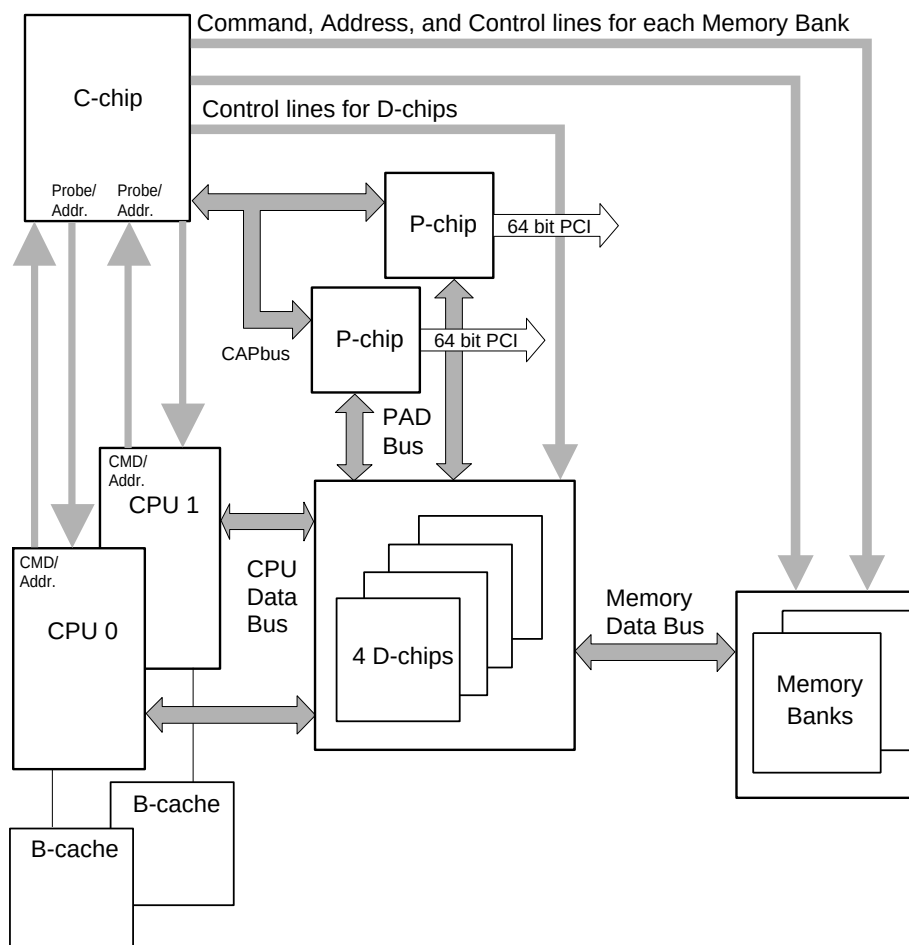
This system is designed to maximize the potential of the Alpha 21264B chip. The traditional bus interconnect has been replaced by a switch-based interconnect system. With a bus design, the processors, memory, and I/O modules must share the bus. As the number of bus users increases, the transactions interfere with one another, increasing latency and decreasing aggregate bandwidth. However, with a switch-based system there is no degradation in performance as the number of CPUs, memory, and I/O users increase. Although the users increase, the speed is maintained.

With a switch-based, or point-to-point interconnect, the performance remains constant, even though the number of transactions multiplies. The switched system interconnect uses a set of complex chips that route the traffic over multiple paths. The chipset consists of one C-chip, four D-chips, and two P-chips,

- **C-chip.** Provides the interface from the CPUs and main memory. Although there is only one in the system, it allows each CPU to do transactions simultaneously.
- **D-chips.** Provide the data path for the CPUs, main memory, and I/O.
- **P-chips.** Provide the interface to two independent 64-bit PCI buses.

The DS20L system supports two CPUs and up to 2 Gbytes memory in 8 DIMM slots. There are two memory banks, each with 4 slots.

The 256-bit memory bus supports two memory banks, yielding a 2.6 Gbyte/sec system bandwidth. Transactions are ECC protected. Upon the receipt of data, the receiver checks for data integrity and corrects any errors. Commands and addresses are parity protected.



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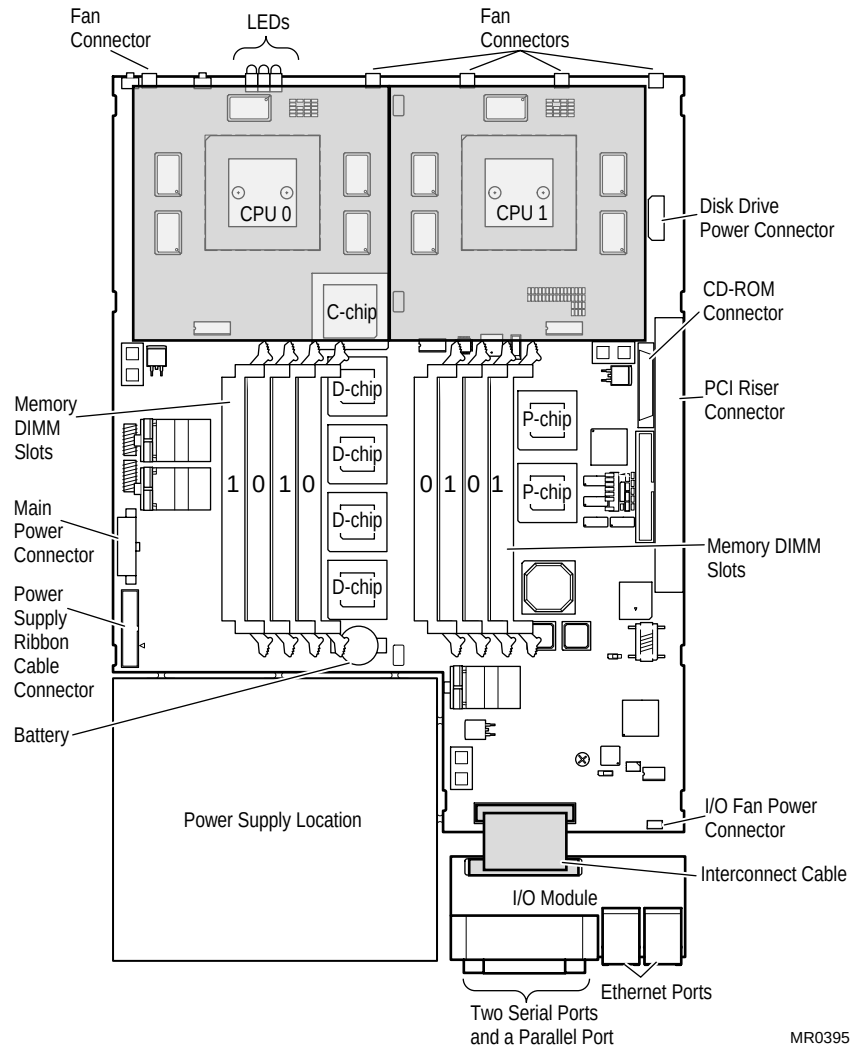
System Block Diagram

## Motherboard

The interconnect switch is implemented on the motherboard by the chipset consisting of one C-chip, two P-chips, and four D-chips. This complex chipset provides the data and address path between the CPUs, memory, and the I/O subsystem.

A flash ROM holds the console code and the NVRAM data.

A single cable connects the I/O daughter card to the motherboard (see lower right of module).



**Component and Connector Locations**

## Memory

The memory bus is 256 bits wide (32 bytes), and the speed is 83 MHz. This yields 2.67 GB/sec bandwidth ( $32 \times 83 \text{ MHz} = 2.6 \text{ GB/sec}$ ).

Memory throughput in this system is maximized by the following features:

- Very low memory latency (120 ns) and high bandwidth with 12 ns clock
- Reliability and availability features, such as ECC memory and command/address parity

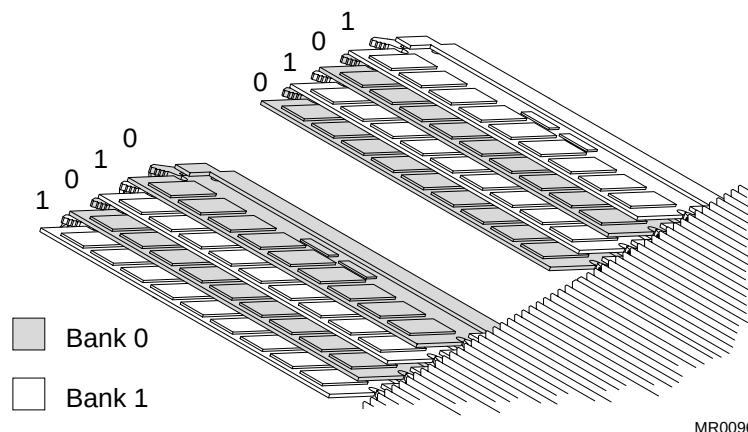
Each memory bank has four slots that accept 168-pin industry-standard DIMMs. System memory can be from 512 Mbytes (4 x 128 MB DIMMs in one bank) to a maximum of 2 Gbytes (both banks filled with 256 MB DIMMs).

### Memory Configuration Rules

- A bank must be fully populated, that is, all four slots in a given bank (0 or 1) *must* be filled.
- Bank 0 must be populated first.
- A bank must use the same type, size, and speed DIMMs. Bank 0 and Bank 1 may have different type, size, and speed of DIMMs from each other, however.

### Memory Interleaving

With two memory banks, memory read-write operations are two-way interleaved. Interleaved operations reduce the average latency and increase the memory throughput over noninterleaved operations.



Memory DIMMs

## System I/O

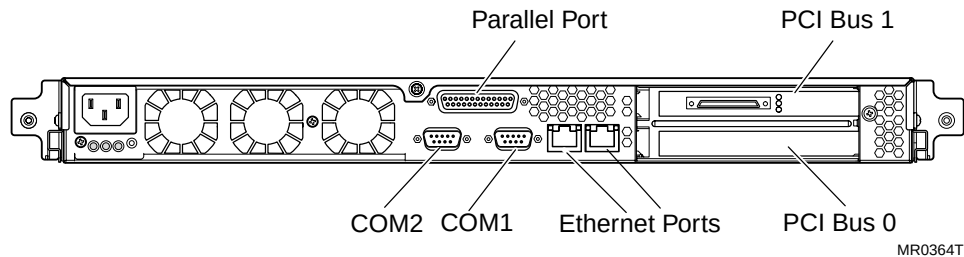
There are two 64-bit PCI buses. Industry-standard PCI I/O buses allow you to use inexpensive, widely available I/O options. Both 32-bit and 64-bit PCI options can be used in the 2/3 length slots (8.375 inches).

### PCI Bus

The industry-standard PCI bus is the number one choice for high-performance I/O options, such as disk storage and high-performance video applications.

The PCI bus implementation has the following characteristics:

- Fully compliant with the PCI Version 2.1 Specification
- Operates at 33 MHz, delivering a peak bandwidth of 500 MB/sec; over 250 Mbytes/sec for each PCI bus
- Supports 1 slot on each bus.
- Supports peer-to-peer I/O operations
- Supports three address spaces: PCI I/O, PCI memory, and PCI configuration space
- Supports byte/word, tri-byte, and longword operations
- Exists in noncached address space only



I/O Ports

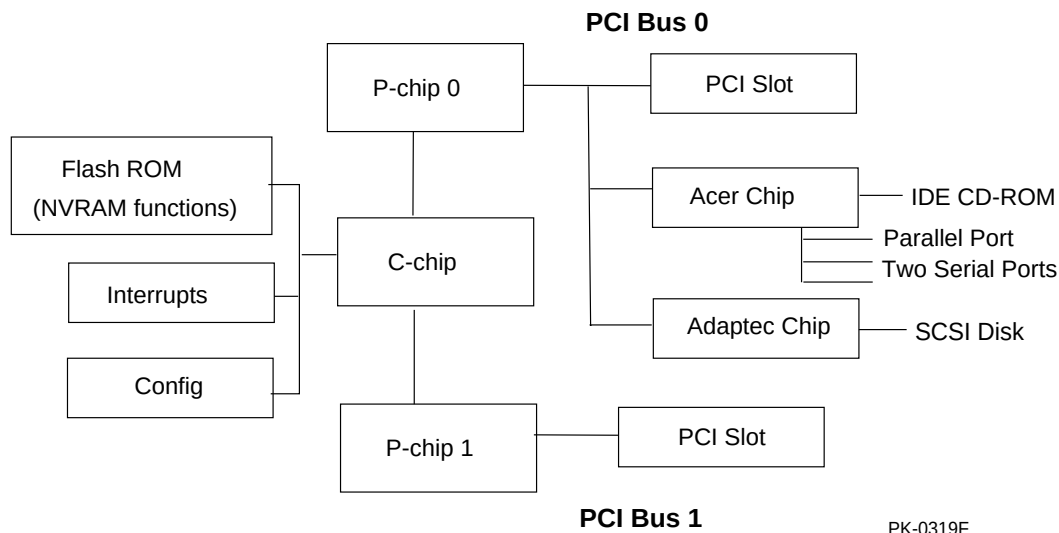
### I/O Implementation

Two separate 64-bit PCI buses are provided, each with one slot. PCI options are installed from the side of the system and plug into the PCI riser card. The integrated SCSI controller is on the PCI riser card. At the rear of the system are connectors offering access to two serial communication ports and one parallel port.

As can be seen in the block diagram, the C-chip controls both PCI control chips.

Each PCI bus has a single option slot. In addition, PCI bus 0 supports the following:

- Acer chip, which provides the integrated controller for the IDE devices and support for the communications ports and the parallel port.
- Adaptec chip, which provides support for the integrated controller for SCSI devices.



Block Diagram of I/O Control

## System Control

At the rear of the system is an On/Off button, and at the front there is a Halt button that can be depressed with a long slender object. Also, the system can be turned on by means of the remote console.

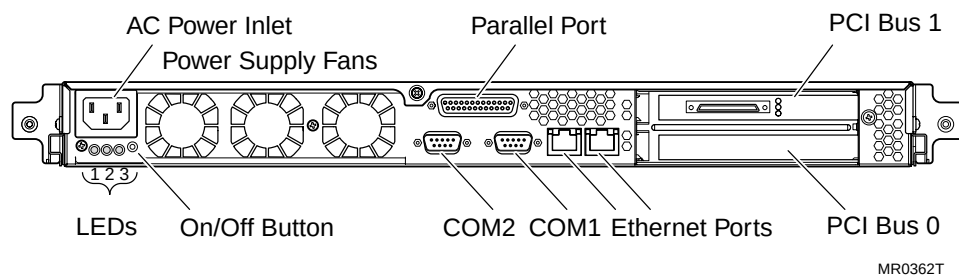
- Environmental monitoring is performed by the Acer chip.

## LEDs

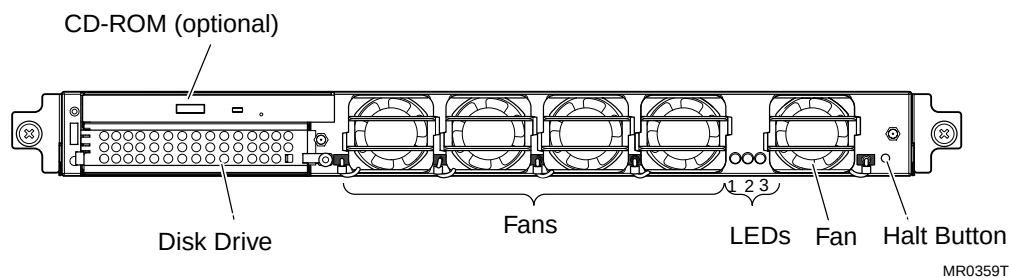
The system has a number of LEDs that indicate system status. System LEDs are on the front and the back of the enclosure, and the two network connectors also have status LEDs.

## Power

The system is powered by a 425 watt DC output power supply. Each has its own power cord that plugs into a power distribution unit in the cabinet. A power distribution unit is required for every eight systems.



Rear of the System



Front of the System

## Storage

The *AlphaServer* DS20L system has an integrated Ultra3 SCSI controller; 18.2, 36.4, and 72.8 Gbyte disk drives are available. An optional 24X CD-ROM drive can be installed (600 Mbytes); the CD uses the IDE controller. One is required per cabinet.

Ultra3 StorageWorks shelves can be installed in the cabinet. Each Ultra3 shelf holds 14 one-inch disks.

## Fibre Channel

Available on these *AlphaServer* systems is the next storage interface, Fibre Channel, which eliminates issues with today's SCSI interfaces such as distance, bandwidth, scalability, and reliability. Fibre Channel (FC) is the answer to not only server-to-storage connections but also to server-to-server networking, because multiple protocols are supported. SCSI, TCP/IP, video, or raw data can all take advantage of high-performance, reliable Fibre Channel technology.

With the KGPSA PCI Fibre Channel adapter, the DS20L systems provide a storage interconnect that is 2.5 times as fast as UltraSCSI: 100 vs 40 Mbytes/sec data throughput. The KGPSA adapter allows you to manage storage including the HSG80 RAID controller in a switched FC topology.

## Server Management

The *AlphaServer* products support important operational and platform management requirements.

### Operational Management

*Server/Network Management. Comaq Insight Manager* is included with every system. This software tool allows you to monitor and control Alpha-based servers. *Insight Manager* consists of two components: a Windows-based console application and server- or client-based management data collection agents. Management agents monitor over 1,000 management parameters. Key subsystems are instrumented to make health, configuration, and performance data available to the agent software. The agents act upon that data, by initiating alarms in the event of faults and by providing updated management information, such as network interface or storage subsystem performance statistics.

### Platform Management

The *AlphaServer* DS20L systems support platform management tasks such as manipulating and monitoring hardware performance, configuration, and errors. For example, the operating systems provide a number of tools to characterize system performance and display errors logged in the system error log file.

In addition, system console firmware provides hardware configuration tools and diagnostics to facilitate quick hardware installation and troubleshooting. The system operator can use simple console commands to show the system configuration, devices, boot and operational flags, and recorded errors. Also, the console aids in inventory support by giving access to serial numbers and revisions of hardware and firmware.

### Security

- Password protection is offered by the SRM console.

## Reliability, Availability, and Maintainability

Reliability and availability features are built into the CPU, memory, and I/O, and implemented at the system level.

### Processor Features

- CPU data cache provides error correction code (ECC) protection.
- Parity protection on CPU cache tag store.
- Multi-tiered power-up diagnostics to verify the functionality of the hardware.

With two processors, when you power up or reset the system, each CPU, in parallel, runs a set of diagnostic tests. If any tests fail, the failing CPU is configured out of the system. Responsibility for initializing memory and booting the console firmware is transferred to the other CPU, and the boot process continues. This feature ensures that a system can still power up and boot the operating system in case of a CPU failure.

### Memory Features

- The memory ECC scheme is designed to provide maximum protection for user data. The memory scheme corrects single-bit errors and detects double-bit errors and total DRAM failure. It also detects RAM address errors.
- Memory failover. The power-up diagnostics are designed to provide the largest amount of usable memory, configuring around errors.

### I/O Features

- ECC protection on the switch interconnect and parity protection on the PCI and SCSI buses.
- Extensive error correction built into disk drives.
- Optional external RAID improves reliability and data security.

### System Features

*Auto reboot.* A firmware environment variable lets you set the default action the system takes on power-up, reset, or after an operating system crash. For maximum system availability, the variable can be set to cause the system to automatically reboot the operating system after most system failures.

*Software installation.* The operating system is factory installed, if a *Tru64 UNIX* system is ordered. The system is also available with no operating system, so that the customer can install Linux.

*Diagnostics.* During the power-up process, diagnostics are run to achieve several goals:

- Provide a robust hardware platform for the operating system by ensuring that any faulty hardware does not participate in the operating system session. This maximizes system uptime by reducing the risk of system failure.
- Enable efficient, timely repair.

The system has a firmware update utility (LFU) that provides update capability for console and PCI I/O adapter firmware.

*Service Tool.* A bit-to-text troubleshooting tool is available to support FRU isolation.

*Thermal management.* The air temperature and fan operation are monitored to protect against overheating and possible hardware destruction. Eleven fans provide cooling; the three fans at the rear cool the power supply.

*Error handling.* Parity and other error conditions are detected on the PCI bus. The memory checking scheme corrects single-bit errors and detects double-bit errors. Multiple ECC corrections to single-bit errors detected by the operating systems help in determining where in the system the error originated. Errors are logged for analysis.

*Uninterruptible power supply.* An external UPS can be used to support critical customer configurations. Because power is maintained for the entire system (CPU, memory, and I/O), power interruptions are completely transparent to users.

## Installation and Maintenance

The systems are designed for easy hardware, software, and option installation. Options ordered with a system are preinstalled and tested at the factory. The *Tru64 UNIX* operating system is also installed at the factory.

When the H9A10 or H9A15 rack cabinet is ordered, the DS20L systems will be factory installed. If any systems are ordered without a cabinet, there is a requirement that any cabinet have a minimum of four systems.

## Clustering

A cluster is a loosely coupled set of systems that behaves (is addressed and managed) like a single system, but provides high levels of availability through redundant CPUs, storage, and data paths. Clusters are also highly scalable; that is, CPU, I/O, storage, and application resources can be added incrementally to efficiently increase capacity. For customers, this translates to reliable access to system resources and data, and investment protection of both hardware and software.

Clustering allows multiple computer systems to communicate over a common interface, share disks, and spread the computing load across multiple CPUs.

For clustered *Tru64 UNIX* systems, TruCluster Software solutions allow users access to network services and provide further failover recovery from server, network, or I/O failures.

## Performance

Hewlett-Packard has an ongoing program of performance engineering, using industry-standard benchmarks that allow comparisons across major vendors' systems. These benchmarks against competitive systems are based on comparable CPU performance, coupled with comparable memory and disk expandability. See Table 1 for the performance numbers of the *AlphaServer DS20L* systems.

System performance, however, is highly dependent upon application characteristics. Thus, benchmark information is one helpful "data point" to be used in conjunction with other purchase criteria such as features, service, and price.

### Sources of Performance Information

Performance information is available on the Internet.

<http://www.compaq.com/alphaserver/performance/index.html>

[http://www.ideasinternational.com/benchmark/spec/specfp\\_s2000.html](http://www.ideasinternational.com/benchmark/spec/specfp_s2000.html)

### Information for Partners

If you are a Channel or Reseller Partner, you can find the tools, resources, and information you need to conduct business online on the secure Compaq Business Partner site:  
<http://bps.compaq.com/businesspartner>

Also see the following:

Compaq Solutions Alliance site at <http://csa.compaq.com>  
and HP's Developer and Solution Partner Portal (DSPP),  
<http://www.hp.com/go/partners>

## Service and Support

Hewlett-Packard provides a comprehensive set of services that range from migration, consulting, and training, to direct support of Alpha systems, software, and applications. For information on services, see  
<http://www.compaq.com/services>  
<http://www.hp.com/hps/>

### Hardware Warranty

The *AlphaServer DS20L* system and components, including CPU, memory, PCI controllers, and power supplies, have a 1-year on-site, 5-day per week, 9-hour per day hardware warranty with next business day response time.

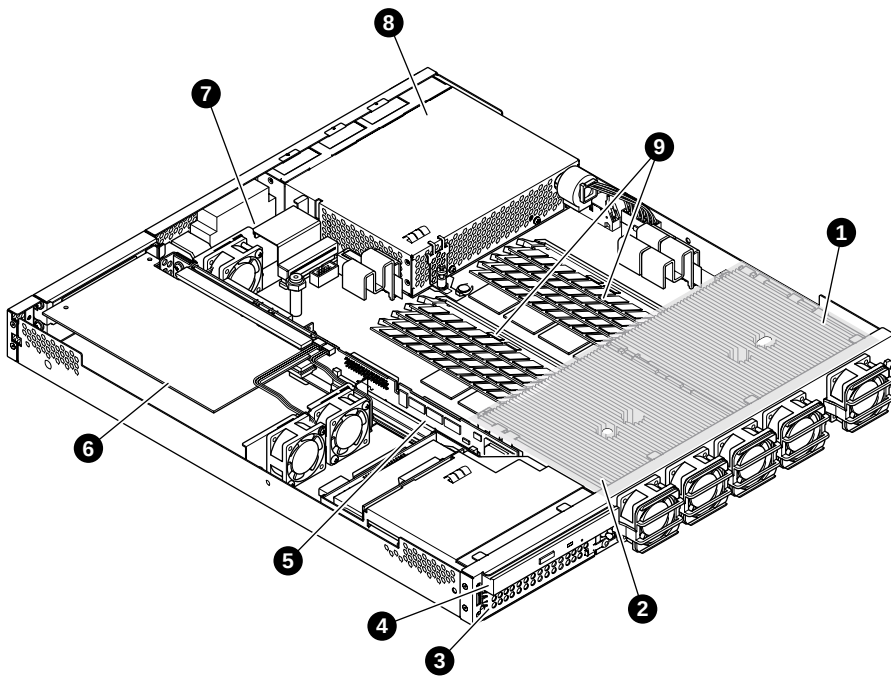
StorageWorks components are supported by the comprehensive StorageWorks warranty: five years for disks, three years for controllers, two years for tape devices, and one year for other components. The first year includes on-site next-day response time. Network products carry the network products warranty.

Users can upgrade to higher levels of service through a variety of hardware supplemental services.

### Software Warranty

The warranty for *Tru64 UNIX* is conformance to SPD with advisory telephone support for a period of 90 days. Users can upgrade to higher levels of service through a variety of software supplemental services.

## hp AlphaServer DS20L System Diagram



1. CPU0
2. CPU1
3. Hard drive
4. CD-ROM  
(optional)
5. PCI riser card
6. PCI options
7. I/O daughter card
8. Power supply
9. Memory DIMMs

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## System Features at a Glance

Table 1 provides a quick reference to features of *AlphaServer* DS20L systems.

**Table 1 AlphaServer DS20L Features**

|                                 |                                                                                           |        |
|---------------------------------|-------------------------------------------------------------------------------------------|--------|
| CPU Features                    | 68/833                                                                                    |        |
| Processor                       | 2 Alpha 21264B chips                                                                      |        |
| CPU clock speed                 | 833 MHz                                                                                   |        |
| Cache on chip                   | 64 KB I-cache                                                                             |        |
|                                 | 64 KB D-cache                                                                             |        |
| On-board cache                  | 4 MB per processor                                                                        |        |
| Internal Storage                |                                                                                           |        |
| Hard disk                       | Ultra3 SCSI drive (18.2, 36.4, or 72.8 GB)                                                |        |
| Removable media                 | Optional CD-ROM drive                                                                     |        |
| Memory (maximum)                | 2 GB                                                                                      |        |
| Performance                     | 1 CPU                                                                                     | 2 CPUs |
| SPECint2000                     | 537                                                                                       | —      |
| SPECfp2000                      | 726                                                                                       | —      |
| SPECfp_rate2000                 | 8.40                                                                                      | 14.3   |
| Linpack 1000 x 1000             | 1232                                                                                      | 2313   |
| Standard Features               | 2 serial ports, 1 parallel port, operating system license, Internet software              |        |
| I/O System                      |                                                                                           |        |
| I/O slots                       | 2 64-bit, 33 MHz 2/3 length slots                                                         |        |
| Maximum PCI throughput          | Over 500 MB/sec with two 256 MB/sec buses                                                 |        |
| High Availability Features      |                                                                                           |        |
| System                          | System auto reboot, thermal management, RAID, memory failover, ECC memory, ECC cache, SMP |        |
|                                 | CPU failover, error logging, optional uninterruptible power supply                        |        |
| Tru64 UNIX TruCluster Solutions | SCSI                                                                                      |        |
| Operating Systems               | Tru64 UNIX and Linux                                                                      |        |
| Warranty                        |                                                                                           |        |
| Hardware                        | 1-year, on-site, 5 day x 9 hour warranty with next business day response time             |        |
| Software                        | 90-day telephone advisory support for Tru64 UNIX                                          |        |

## Physical Characteristics

Table 2 details basic physical characteristics of the system.

**Table 2 AlphaServer DS20L Physical Characteristics**

| Dimensions                                                                                                  | DS20L System                                                                    | H9A10 M-Series cabinet                                                                                                               | H9A15 M-Series cabinet                    |
|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Height x width x depth                                                                                      | 1.75 x 17.5 x 20.5 in.<br>4.5 x 44.5 x 52 cm                                    | 67 x 23.6 x 41.5 in.<br>170 x 60 x 100 cm                                                                                            | 79 x 23.6 x 41.5 in.<br>200 x 60 x 100 cm |
| Weight                                                                                                      | 9.4 kg (21 lbs)                                                                 | Configuration dependent                                                                                                              | 1,000 lbs/450 kg                          |
| <b>Environmental</b>                                                                                        |                                                                                 |                                                                                                                                      |                                           |
| Temperature                                                                                                 | Operating<br>Nonoperating<br>Storage (60 days)<br>Rate of change<br>(operating) | 50 – 95° F (10 – 35° C)<br>41 – 122° F (5 – 50° C)<br>–40 to 151° F (–40 to 66° C)<br>20° F /hr / 11° C /hr                          |                                           |
| Relative Humidity                                                                                           | Operating<br>Nonoperating<br>Storage (60 days)                                  | 10 – 90 % (maximum wet bulb: 82° F/28° C)<br>10 – 95 % (maximum wet bulb: 90° F/32° C)<br>10 – 95 % (maximum wet bulb: 115° F/46° C) |                                           |
| Maximum altitude                                                                                            | Operating<br>Nonoperating                                                       | 10,000 ft (3,048 m)<br>40,000 ft (12,192 m)                                                                                          |                                           |
| Shock                                                                                                       | Operating                                                                       | 10 G, 10ms half-sine                                                                                                                 |                                           |
| Vibration                                                                                                   | Operating<br><br>Nonoperating                                                   | 5 to 500 Hz at 0.1 G peak (limited by 0.02"/0.5 mm double amplitude)<br>1.03 Grms 5–300 Hz                                           |                                           |
| <b>Electrical</b>                                                                                           |                                                                                 |                                                                                                                                      |                                           |
| Nominal voltage range (Vac)                                                                                 | 200–240                                                                         |                                                                                                                                      |                                           |
| Voltage range (Vac)                                                                                         | 180–250                                                                         |                                                                                                                                      |                                           |
| Power source phase                                                                                          | Single                                                                          |                                                                                                                                      |                                           |
| Nominal frequency (Hz)                                                                                      | 50/60                                                                           |                                                                                                                                      |                                           |
| Frequency range (Hz)                                                                                        | 59–61/49–51                                                                     |                                                                                                                                      |                                           |
| Single power supply (amps)                                                                                  | 1.4 nominal @ 240 VAC                                                           |                                                                                                                                      |                                           |
| Power consumption                                                                                           | 275 watts, typical configuration                                                |                                                                                                                                      |                                           |
| Heat dissipation                                                                                            | 939 BTU/hr                                                                      |                                                                                                                                      |                                           |
| <b>Acoustics—Declared values per ISO 9296 and ISO 7779</b>                                                  |                                                                                 |                                                                                                                                      |                                           |
| Current values for specific configurations are available from Hewlett-Packard representatives. 1 B = 10 dBA |                                                                                 |                                                                                                                                      |                                           |
| Acoustics                                                                                                   | L <sub>WAd</sub> , B                                                            | L <sub>pAm</sub> , dBA (bystander position)                                                                                          |                                           |
| Idle                                                                                                        | 6.7                                                                             | 51                                                                                                                                   |                                           |
| Operating                                                                                                   | 6.7                                                                             | 51                                                                                                                                   |                                           |



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